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Mediators of Social Acceptance Among Emerging Adult Survivors of Childhood Cancer

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Purpose: Examine associations of social developmental factors (e.g., peer/parent social attachment, romantic relationships) and perceptions of social acceptance among emerging adult survivors of childhood cancer.
Methods: A cross-sectional, within-group design was used. Questionnaires included the Multidimensional Body-Self Relations Questionnaire, Inventory of Parent and Peer Attachment, Adolescent Social Self-Efficacy Scale, Personal Evaluation Inventory, Self-Perception Profile for Adolescents, and demographics. Correlations were utilized to determine associations between general demographic, cancer-specific, and the psychosocial outcome variables. Peer and romantic relationship self-efficacy were assessed as potential mediators of social acceptance in three mediation models. Relationships between perceived physical attractiveness, peer attachment, parental attachment, and social acceptance were assessed.
Results: Data were collected from $N = 52$ adult participants ($M_{age} = 21.38$ years, standard deviation = 3.11 years) diagnosed with cancer as a child. The first mediation model demonstrated a significant direct effect of perceived physical attraction on perceived social acceptance and retained significance after adjusting for indirect effects of the mediators. The second model demonstrated a significant direct effect of peer attachment on perceived social acceptance; however, significance was not retained after adjusting for peer self-efficacy, suggesting the relationship is partially mediated by peer relationship self-efficacy. The third model demonstrated a significant direct effect of parent attachment on perceived social acceptance; however, significance was not retained after adjusting for peer self-efficacy, suggesting the relationship is partially mediated by peer self-efficacy.
Conclusion: Relationships between social developmental factors (e.g., parental and peer attachment) and perceived social acceptance are likely mediated by peer relationship self-efficacy in emerging adult survivors of childhood cancer.

Keywords: cancer survivor, emerging adulthood, social acceptance

Introduction

Emerging adulthood, defined as 18 through 25 years of age, is recognized as a distinct period of personal growth,[1] typically with central foci on social functioning and identity development.[2] However, limited research has focused on emerging adult survivors of childhood cancer,[3] a group that experiences multiple psychosocial challenges[4,5] and chronic health conditions.[6,7] Perceptions of social acceptance (i.e., fitting in among acquaintances/friends) is a salient social factor for this group,[1] as emerging adult survivors of childhood cancer are particularly vulnerable to deficits in social functioning as compared with peers without a history of cancer,[8,9] which is important for identity

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development and achieving social goals.[10] Although a range of factors influence social functioning, pertinent areas include peer and romantic relationship self-efficacy, attachment to peers and parents, and perceptions of physical attractiveness.[11–13]

As compared with peers without a history of cancer, cancer survivors are affected by social isolation and engage in fewer social activities,[14] which may be related to poorer perceptions of self-worth and increased negative perceptions of physical attractiveness. During adolescence and young adulthood, body image and self-perceived physical attractiveness may be linked to perceiving oneself and others as more acceptable among peer groups, suggesting negative perceptions of physical attractiveness may have long-term detrimental effects on social development for emerging adult cancer survivors, particularly those with delayed puberty and physical issues (e.g., alopecia, obesity, amputation).[15]

Among emerging adults who have experienced cancer, social challenges associated with peer and romantic relationships have been reported and are conceptualized as distinct from other types of relationships.[16] Despite the importance of these relationships, little is known regarding peer and romantic relationship self-efficacy among emerging adult survivors of childhood cancer or the impact of self-efficacy on perceived social acceptance.

The conceptual framework for the study is informed by tenets of social cognitive theory,[10] as well as the broader emerging adulthood literature.[2] Social cognitive theory proposes that an individual's knowledge acquisition and sustained behaviors are directly related to observing others in social experiences and requires interactions in a social learning context with dynamic interaction between personal, behavioral, and environmental factors.[10] Based on previous research in emerging adult cancer survivors,[17] self-efficacy, or confidence in one's ability to perform a behavior to achieve an outcome,[18] was operationalized as peer and romantic relationship self-efficacy for this study. Emerging adult literature suggests that improving self-worth, reducing doubt, and exposure to romantic and peer experiences can promote relational competence and enhance perceptions of social acceptance.[19,20]

Parental and peer attachment[21] and perceived physical attractiveness[15] were selected as additional personal and environmental factors that may influence the outcome of perceived social acceptance. We sought to explore the potential impact of self-efficacy (i.e., peer and romantic relationship self-efficacy) and personal and environmental barriers and facilitators (i.e., perceptions of physical attractiveness and attachment to parents and peers) on a perceived social acceptance. Specifically, peer and romantic relationship self-efficacy were assessed as potential mediators of perceived social acceptance in three mediation models.

The purpose of this study was to examine (1) psychosocial determinants of perceived social acceptance that have been investigated in noncancer populations, including physical attractiveness and peer attachment, and (2) peer and romantic relationship self-efficacy as mediators of the associations between these predictors and perceived social acceptance in emerging adult survivors of childhood cancer. It was hypothesized that parent attachment, peer attachment, and perceived physical attractiveness would independently be related to perceived social acceptance.

Methods

Study design

A cross-sectional, within-group design examined social developmental factors as salient features related to perceived social acceptance among emerging adult survivors of childhood cancer. Relationships among perceived physical attractiveness, peer attachment, parental attachment, and perceived social acceptance were assessed. Mediators of these associations included peer relationship self-efficacy and romantic relationship self-efficacy. A sample of eligible emerging adults who were diagnosed with pediatric cancers between the ages of 5 and 19 were recruited for participation through the VCU Medical Center Pediatric Hematology/Oncology Outpatient Clinic. All aspects of this study were approved by the VCU Institutional Review Board (IRB) and the Massey Cancer Center Protocol Review and Monitoring System Committee. All participants provided written informed consent. Participants completed questionnaires through paper and pencil or online. Each participant received a $20 gift card as compensation for participation. The study was approved by the Institutional Review Board.

Participants

Initial contact to identify eligible emerging adults was made by staff at the clinic using patient records. Eligible participants were contacted in person during regularly scheduled follow-up visits at the clinic and/or were sent informational packets through mail describing the study in detail. An initial welcome letter, written at the sixth grade reading level, explained the study to the participant. This letter included all information provided in a standard consent form (description of the study, procedures, participant compensation, risks/benefits, confidentiality, voluntary participation, the ability to withdraw at any time, the ability to not answer any question(s) of their choosing, how to contact the researchers for general questions, and/or therapy referral list and contact information for the VCU Office of Human Subjects Protection). A return form with prepaid return envelope was provided to eligible participants as well. In addition to requesting participant contact information, the return form asked participants to choose whether they would like to participate through paper and pencil packets or online survey system.

Consent was implied by the return form participants completed stating that they would like to be a part of the study. VCU Institutional Review Board (IRB) regulations state that this type of implied/waived consent was appropriate because the study involved no more than minimal risk and that a detailed descriptive letter was sent to fully explain the study. Additionally, IRB guidelines for online data collection state that consent is implied in that if the participants complete the questionnaires they have, in turn, provided consent. Reasons for nonparticipation was not recorded. Participant eligibility included: (1) current age of 18 to 25 years, (2) completion of cancer treatment(s) ≥ 6 months before participation, (3) no history of premorbid neurological impairment, and (4) English-language proficiency.

Instrumentation

Demographics. A questionnaire assessed demographic and clinical information. Basic demographic information and

disease-specific information was obtained through patient medical records for all eligible participants (e.g., diagnosis, cancer staging, treatment modalities).

Treatment intensity. The Intensity of Treatment Rating Scale 2.0 was used to assess treatment intensity using patient records.[22] A pediatric oncology nurse practitioner indicated a level of treatment intensity ranging from 1 (*least intensive treatments*) to 4 (*most intensive treatments*) using deidentified treatment-related factors such as diagnosis, stage or risk level, relapse status, and whether or not surgery, chemotherapy, radiation therapy, and/or transplant were used as treatment modalities. Content validity between criterion and external raters was excellent ($r = 0.95$). Interrater reliability was excellent as well ($r = 0.87$).

Physical attractiveness. The Appearance Evaluation Subscale (7 items) of the Multidimensional Body-Self Relations Questionnaire (MBSRQ)[23] was used to assess perceptions of physical attractiveness. Internal reliability of the MBSRQ subscales has been reported from 0.70 to 0.91 with 1-month test/retest reliability ranging from 0.71 to 0.94. This measure has been used to assess college students ages 18 to 23,[24] multiracial populations ages 18 to 30,[25] and adult cancer survivors.[26,27] Cronbach's α for this study was 0.88.

Attachment to parents and peers. Participants' relations with their primary caregivers and peers were examined using the Inventory of Parent and Peer Attachment (IPPA).[28] Normed on college students ages 16 to 20, the IPPA has 28 items that make up a global parental attachment scale. The subscales of the IPPA have a Cronbach's $\alpha = 0.91$.[28] For this study, Cronbach's α was 0.90 for parent communication, 0.93 for parent trust, and 0.87 for parent alienation. Cronbach's α for the overall measure was 0.87. The global peer attachment scale has 25 items with a Cronbach's α reported at 0.87.[28] For this study, Cronbach's α was 0.90 for peer communication, 0.93 for peer trust, and 0.62 for peer alienation. Cronbach's α for the overall measure was 0.81.

Peer relationship self-efficacy. The Friendship/Intimacy (7 items) and Social Groups/Parties (5 items) subscales of the Adolescent Social Self-Efficacy Scale[29] were used to examine peer relationship self-efficacy. This scale was originally normed on high school students (ages 13–19). Wording was modified slightly to better reflect the emerging adult age group under investigation. For example, ''boy and girl'' was changed to ''male and female,'' ''kid'' was changed to ''peer,'' and ''recess'' was changed to ''leisure time.'' This test has demonstrated adequate psychometric properties: Cronbach's α of 0.90–0.92 and a test/retest of 0.84 across both sexes. The combined Cronbach's α for these two subscales, as utilized in the current study, was 0.94.

Romantic relationship self-efficacy. The Romantic Relationships subscale (7 items) of the Personal Evaluation Inventory[30] was used to assess perceptions of romantic relationship self-efficacy. This scale was normed on college-age students with Cronbach's α ranges from 0.74 to 0.89 for females and 0.67 to 0.86 for males. One-month test/retest scores for the entire scale were 0.80 for females and 0.93 for males. Cronbach's α was 0.91 for the current study.

Social acceptance. The Social Acceptance subscale (5 items) of Self-Perception Profile for Adolescents was utilized to assess perceptions of social acceptance.[31] For the purposes of this study, the stem of each item was modified slightly for emerging adults so participants could provide an assessment of their perceptions of how they believe others view them in addition to their self-perceptions. Internal consistency for the Social Acceptance subscale has been reported at Cronbach's $\alpha = 0.76$.[32] For the current study, Cronbach's α was 0.87 for perceived self-acceptance and 0.81 for how participants perceived they are viewed by others.

Statistical approach

Correlations were utilized to determine associations between general demographic, cancer-specific variables, and the psychosocial outcome variables under investigation (i.e., peer relationship self-efficacy, romantic relationship self-efficacy, and perceived social acceptance). Variables found to be significantly related to outcome variables were then entered into multiple mediation models. The mediation analyses were completed to assess the potential indirect effects of two or more mediating variables concurrently.[33] The analyses included a bootstrapping resampling strategy to determine the significance of the model and specific mediators. One thousand bootstrap samples were utilized. Independent sample *t*-tests and chi-square analyses were used to assess for significant differences between those choosing to participate and those choosing not to participate to provide a more detailed assessment of sample differences that may be related to a person's choice not to engage in such a study. All analyses were conducted using SAS 9.4 (SAS Institute, Inc.). *p*-Values <0.05 were considered statistically significant for all analyses.

Results

Fifty-two emerging adults returned completed questionnaire packets (28 males, 24 females; mean age (M) = 21.38 years, standard deviation (SD) age = 3.11 years) out of 133 potential participants (39.0% participation rate). All participants had been diagnosed with cancer ($M_{age} = 12.59$ years, $SD_{age} = 4.57$ years) and were off active treatment ($M_{time} = 7.32$ years, $SD_{age} = 4.46$ years, range = 6 months to 17.83 years). Age of diagnosis was ($M_{age} = 12.6$ years; $SD_{age} = 4.6$, range = 3–23 years). Most participants were White/Caucasian (86%). The majority reported being single (81%), and over half (58%) lived with one or both parents. Fifty-two percent were in school at the time of participation, with 38% having obtained at least some college education. Diagnoses included leukemias (31%), lymphomas (37%), solid tumors/sarcomas (19%), and central nervous system tumors (13%). See Table 1 for full demographic and treatment information.

Correlations

Demographic and clinical correlations. Psychosocial variable means scores, standard deviations, and ranges are presented in Table 2. Demographic and cancer-specific variables, including treatment intensity, were not related to perceived physical attractiveness or parent or peer attachment (Table 3). Demographic and cancer-specific variables were not associated with psychosocial variables of interest.

Table 1. Participant Demographics

Characteristic	n *(%)*
Gender	
Male	28 (54)
Female	24 (46)
Race/ethnicity	
Caucasian/White	45 (86)
African American	4 (8)
Hispanic/Latino	1 (2)
Asian American	1 (2)
Biracial/multiracial	1 (2)
Marital status	
Single	42 (81)
Married	5 (9)
Partnered	4 (8)
Other	1 (2)
Living arrangement	
With both parents	20 (39)
With mother only	10 (19)
With spouse or partner	6 (12)
Alone	5 (10)
With roommate(s)	10 (19)
Other	1 (2)
In school at time of participation	27 (52)
Highest education level attained	
Some high school	3 (6)
High school/high school equivalent	11 (21)
Some college	27 (52)
Associate's degree	2 (4)
Bachelor's degree	8 (15)
Professional/medical degree	1 (2)
Diagnosed with any learning/attention disorder	9 (17)
Diagnostic categories	
Leukemias	16 (31)
Acute lymphoblastic leukemia	15 (29)
Chronic myelogenous leukemia	1 (2)
Lymphomas	19 (37)
Hodgkin lymphoma	13 (25)
Non-Hodgkin lymphoma	6 (12)
Central nervous system tumor	7 (13)
Medulloblastoma	3 (6)
Germinoma	2 (4)
Hypothalamic glioma	1 (2)
Other	1 (2)
Sarcomas/solid tumors	10 (19)
Osteosarcoma	1 (2)
Ewing's sarcoma	3 (6)
Rhabdomyosarcoma	3 (6)
Neuroblastoma	1 (2)
Testicular/germ cell tumor	2 (4)
Treatment modalities received	
Surgery	32 (62)
Chemotherapy	50 (96)
Radiation	25 (48)
Brain radiation	5 (10)
Bone marrow/stem cell transplant	2 (4)
No. of treatment modalities received	
One modality	14 (27)
Two modalities	18 (34)
Three modalities	16 (31)
Four modalities	4 (8)

(*continued*)

Table 1. (Continued)

Characteristic	n *(%)*
Treatment Intensity	
Moderately intensive treatments	21 (40)
Very intensive treatments	27 (52)
Most intensive treatments	4 (8)
Experienced relapse or secondary cancer	4 (8)
Perceived likelihood of relapse	
Very likely	3 (6)
Likely	4 (8)
Unlikely	13 (25)
Very unlikely	19 (36)
I have not thought about this question	13 (25)
Diagnosed with other chronic illness(es)	12 (23)
Diagnosed with other acute illness(es)	9 (17)
Diagnosed with any mental illness(es)	8 (15)

$N = 52$.

Psychosocial variable correlations

Physical attractiveness. Increased perceived physical attractiveness was related to greater peer ($r = 0.50$, $p < 0.001$) and romantic relationship self-efficacy ($r = 0.45$, $p < 0.001$). Those who felt more physically attractive perceived themselves as more socially accepted ($r = 0.58$, $p < 0.001$) and believed their peers perceived them as being more socially acceptable ($r = 0.55$, $p < 0.001$).

Parental attachment. Participants who were more attached to their parents reported feeling more attached to their peers ($r = 0.49$, $p < 0.001$), having greater peer relationship self-efficacy ($r = 0.43$, $p < 0.001$), and perceived themselves as more socially accepted ($r = 0.42$, $p < 0.01$).

Peer attachment. Those with greater peer attachment reported greater peer relationship self-efficacy ($r = 0.59$, $p < 0.001$), perceived themselves as more socially accepted ($r = 0.53$, $p < 0.001$), and believed their peers viewed them as more socially acceptable ($r = 0.45$, $p < 0.001$).

Table 2. Psychosocial Variable Mean Scores, Standard Deviations, and Ranges

Variable	M	*SD*	*Range*
Physical attractiveness	24.74	6.15	9.00–35.00
Parental attachment	103.30	23.95	47.00–137.00
Peer attachment	94.68	17.55	45.00–120.00
Peer relationship self-efficacy	63.07	16.20	12.00–84.00
Romantic relationship self-efficacy	18.22	6.26	7.00–28.00
Social acceptance—self	15.26	3.69	5.00–20.00
Social acceptance—other	15.94	2.91	9.00–20.00
Social acceptance—health peer	15.64	2.19	11.00–20.00
Social acceptance—peer w/cancer	14.94	2.83	9.00–20.00

$N = 52$, M = mean
SD, standard deviation.

Table 3. Intercorrelations Among Psychosocial Variables and Demographic Variables

Variable	*Peer self-efficacy*	*Romantic self-efficacy*	*Social acceptance—self*	*Physical attractiveness*	*Parent attachment*	*Peer attachment*
Gender	–0.13	–0.08	–0.16	–0.11	0.05	0.13
Current age	0.09	–0.07	–0.06	–0.12	0.07	0.05
Marital status	0.09	0.34	0.25	0.10	0.26	0.18
Living arrangement	0.13	0.39	0.27	0.37	0.28	0.25
Learning disability	–0.05	–0.17	–0.16	–0.25	–0.16	–0.24
Diagnosis	0.08	0.20	0.13	0.29	0.25	0.21
Age at diagnosis	–0.07	0.13	0.09	–0.05	–0.11	0.14
Other chronic illnesses	–0.18	–0.26	–0.21	–0.28	–0.09	–0.10

$N=52$; Pearson correlations calculated for analyses with two continuous variables or one continuous and one dichotomous variable, eta correlation ratios for analyses with a nominal and a continuous variable.

Peer and romantic relationship self-efficacy. Greater peer relationship self-efficacy was related to greater romantic relationship self-efficacy ($r=0.63$, $p<0.001$). Those with higher peer and romantic relationship self-efficacy also reported feeling more socially accepted (peer: $r=0.71$, $p<0.001$, romantic: $r=0.55$, $p<0.001$) and were more likely to believe their peers find them socially acceptable (peer: $r=0.65$, $p<0.001$, romantic: $r=0.41$, $p<0.01$).

Participants versus nonparticipants

Supplementary Table S1 include demographic comparisons of participants vs. nonparticipants. Results of the independent sample *t*-tests (Supplementary Table S2) indicated that participants and nonparticipants were not significantly different with respect to age at diagnosis ($t(143)=-1.61$, $p=$ns), time since treatment ended ($t(121)=0.96$, $p=$ns), current age ($t(145)=0.24$, $p=$ns), treatment intensity ($t(120)=1.37$, $p=$ns), length of treatment ($t(118)=0.48$, $p=$ns), or gender ($t(145)=-0.42$, $p=$ns). Those who participated were somewhat more likely to have undergone surgery than those who did not ($t(139)=-2.13$, $p=0.05$). There were no participation differences based on whether or not the patient received chemotherapy ($t(139)=0.33$, $p=$ns), radiation ($t(139)=-1.23$, $p=$ns), brain radiation ($t(139)=-1.57$, $p=$ns), or bone marrow/stem cell transplant ($t(139)=1.14$, $p=$ns).

There were no significant differences in relapse status among those who participated and those who did not ($t(140)=-1.22$, $p=$ns). Nonparticipants were more likely to be of a minority background than those who chose to participate ($\chi^2=14.21$, $p<0.01$). However, there was no significant difference in cancer diagnosis with respect to those who participated and those who did not ($\chi^2=3.75$, $p=$ns).

Multiple mediation models

Peer and romantic relationship self-efficacy were assessed as potential mediators of perceived social acceptance (Fig. 1).

The first model included perceived physical attractiveness as the predictor variable, perceived social acceptance as the criterion, and both peer and romantic relationship self-efficacy as potential mediating variables. First, perceived social acceptance was regressed on the predictor variable (perceived physical attractiveness). The relationship was significant, $\beta=0.96$, $p=0.001$. Second and third equations were analyzed. In the second equation, the mediators (romantic relationship self-efficacy and peer relationship self-efficacy) were regressed on the predictor variable perceived physical attractiveness. The results indicated there was a significant relationship between romantic relationship self-efficacy and perceived physical attractiveness, $\beta=0.46$, $p=0.0008$, and a significant relationship between peer relationship self-efficacy and perceived physical attractiveness, $\beta=1.3$, $p=0.0002$. The third equation involved regressing the outcome variable (perceived social acceptance) simultaneously on perceived physical attractiveness and romantic relationship self-efficacy and peer relationship self-efficacy.

The model was statistically significant ($p=0.001$). The indirect effect of peer relationship self-efficacy ($p=0.0003$) was significant and romantic relationship self-efficacy was not ($p=0.40$). See Figure 1 for depiction of model effects.

The second model included peer attachment as the independent variable, perceived self-perceptions of social acceptance as the dependent variable, and peer and romantic relationship self-efficacy as potential mediating variables. The total effect of peer attachment on perceived social acceptance was significant, $\beta=2.5$, $p<0.0001$. Second and third equations were analyzed. In the second equation, the mediators (romantic relationship self-efficacy and peer relationship self-efficacy) were regressed on peer attachment. The results indicated a significant relationship between romantic relationship self-efficacy and peer attachment, $\beta=0.09$, $p=0.05$, as well as a significant relationship between peer relationship self-efficacy and peer attachment, $\beta=0.54$, $p<0.0001$. The third equation involved regressing the outcome variable (perceived social acceptance) simultaneously on peer attachment and romantic relationship self-efficacy and peer relationship self-efficacy and was insignificant ($p=0.10$). The indirect effect of peer relationship self-efficacy ($p=0.003$) was significant and romantic relationship self-efficacy was nonsignificant ($p=0.11$). See Figure 1.

The third model included parental attachment as the independent variable, perceived social acceptance as the dependent variable, and both peer and romantic relationship self-efficacy as potential mediating variables. The total effect of parental attachment on perceived social acceptance was significant, $\beta=2.7$, $p=0.002$. Second and third equations were analyzed. In the second equation, the mediators (romantic relationship self-efficacy and peer relationship self-efficacy) were regressed on parent attachment. The results

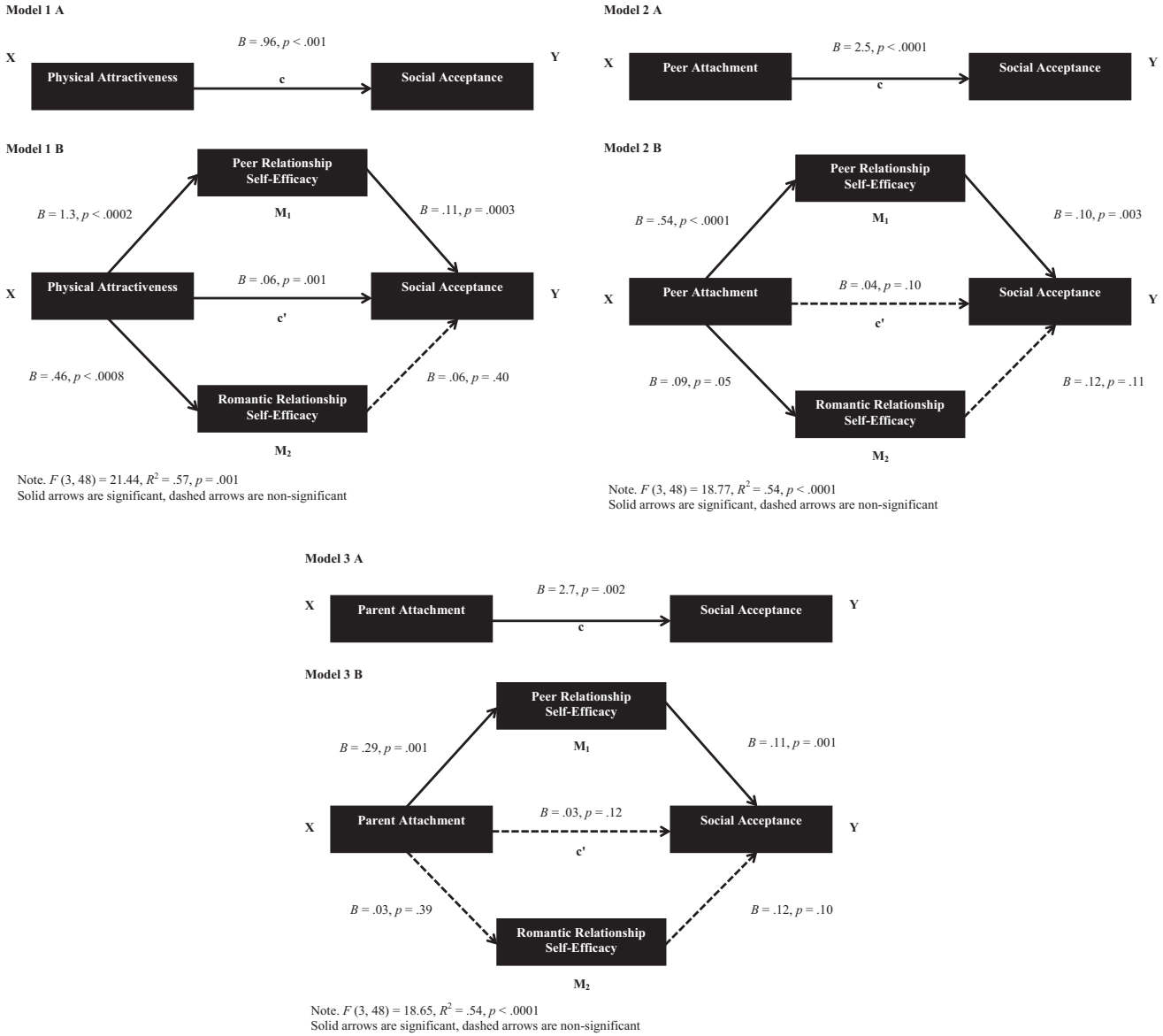

FIG. 1. Multiple mediation models.

indicated there was a nonsignificant relationship between romantic relationship self-efficacy and parental attachment, $\beta = 0.03$, $p = 0.39$, and a significant relationship between peer relationship self-efficacy and parental attachment, $\beta = 0.08$, $p = 0.001$. The third equation involved regressing the outcome variable (perceived social acceptance) simultaneously on parental attachment and romantic relationship self-efficacy and peer relationship self-efficacy.

Results indicated no relationship between parental attachment and perceived social acceptance ($p = 0.12$). The indirect effect of peer relationship self-efficacy ($p = 0.001$) was significant, but romantic relationship self-efficacy was not ($p = 0.10$). See Figure 1 for depiction of model effects.

Discussion

This study explored constructs central to social cognitive theory (i.e., self-efficacy and personal and environmental factors) to identify positive social skill development and adaptation throughout emerging adulthood among survivors of childhood cancer. Peer and romantic relationship self-efficacy were assessed as potential mediators of perceived social acceptance in three mediation models. It was hypothesized that parent attachment, peer attachment, and perceived physical attractiveness would independently be related to perceptions of social acceptance. Demographic and cancer-specific variables were not associated with psychosocial variables of interest. In multiple mediation model testing, peer relationship self-efficacy mediated the relationship between peer attachment and parental attachment on perceived social acceptance. Although, romantic relationship self-efficacy did not mediate the relationship between perceived physical attraction on perceived social acceptance, peer relationship self-efficacy was a mediator of the relationship between physical attractiveness and perceived social acceptance.

Among individuals who have experienced a cancer diagnosis, social challenges related to peer and romantic relationships are robust.[34–39] The current literature exploring

romantic relationships and social acceptance among adolescents and emerging adults suggests that building self-esteem and reducing self-doubt related to social relationships can positively impact perceptions of social acceptance, including how individuals view themselves and how individuals believe others view them.[20] Furthermore, romantic experience relates to social acceptance and friendship and romantic competence.[19] Altogether, these articles suggest that gaining confidence in various aspects of social domains (e.g., romantic relationships) may relate more broadly to perceived social acceptance. Therefore, we hypothesized that romantic relationship self-efficacy among emerging adult survivors of childhood cancer may be related to perceived social acceptance. While some studies have explored these social challenges, little information is available concerning peer and romantic relationship self-efficacy among emerging adult survivors of childhood cancer.

Previous attachment studies have demonstrated the importance of secure parent and peer attachments in the development of adaptive social outcomes.[34,40] In this study, peer relationship self-efficacy mediated the relationship between parental attachment and perceived social acceptance, aligning with prior research. Thus, the ability to develop peer relationship self-efficacy may have an important role in informing social connections and perceptions of acceptance across the cancer survivorship trajectory. Our findings indicated that evidence-based interventions incorporating social cognitive activities that target peer self-efficacy, such as social skills groups, young adult semistructured support groups focusing on peer relationships and perceived social acceptance, peer modeling, and cognitive interventions (e.g., cognitive behavioral therapy; acceptance and commitment therapy), are needed for emerging adult survivors of childhood cancer.[39,41]

Although limited intervention-based literature exists, there is emerging evidence that hospital-, school-, and camp-based interventions focused on enhancing social skills, identity development, building peer relationship and leadership skills, and promoting social interactions while reducing social isolation, all may support the development of social competence, overall social functioning,[35,41] and higher life satisfaction. It has been recommended that such interventions incorporate a focus on addressing concerns with executive functioning and other cognitive late effects that may impact social skill development as well.[42]

The impact of parent attachment on perceived social acceptance was fully mediated by peer relationship self-efficacy. Thus, future emerging adult studies should explore the role of peer and parental dynamics in adaptation throughout treatment and during survivorship. For example, a better understanding of attachment styles and support systems between parent/patient dyads may provide important information relating to confidence in peer and romantic relationships and perceptions of social acceptance.[35]

Peer relationship self-efficacy mediated the relationship between perceived physical attractiveness and perceived social acceptance. This is consistent with previous research suggesting individuals who are physically attractive may also be viewed as being more popular or socially accepted among peer groups.[43–47] The processes that determine how physically attractive emerging adult survivors of childhood cancers feel should be further explored within the context of social and relationship development. Although multiple forums may be explored, this critical emerging adult development may be promoted in the context of individual- and/or group-based evidence-based interventions. This may be accomplished through in-person or telehealth formats targeting a number of salient aspects of emerging adult life, including dating, physical and emotional intimacy, body image/physical attractiveness and physical late effects, disclosing one's cancer history, fertility concerns, and enhancing emotional maturity.[37,38]

This study is not without limitations. The relatively small number of participants restricts the detection of medium to large effect sizes. Further, the small sample limits the generalizability of the results. The sample size might have been a function of a relatively low response rate (39%). A common primary challenge for conducting research with adolescents and young adults with cancer is a low participation rate. Although not ideal, this rate was consistent with previous survey studies in this population.[48–50] Furthermore, there is limited ability to generalize findings of this study to other cancer populations or other emerging adults who have experienced chronic and/or life-threatening illnesses. As with most cross-sectional studies, emerging adult survivors of pediatric cancers who met eligibility criteria self-selected whether to participate, creating potential problems with selection bias. We sent information regarding the study to all eligible participants through mail and recruited in the hematology/oncology clinic in an effort to promote recruitment opportunities.

Although these efforts were made, recruitment of participants from diverse backgrounds was quite poor (Supplementary Table S1). This limits generalizability of findings as our interpretation of results may be less likely to represent the experiences of all survivors of childhood cancer across backgrounds. Future studies would benefit from determining methods of better enhancing inclusion of participants from diverse backgrounds. In addition to recruiting electronically (e.g., email, other forms of social media), which is likely the preferred method for emerging adults in general, including emerging adult survivors as key stakeholders when developing future studies may also lead to improved recruitment of individuals from diverse backgrounds. While reliability data for all measures were available, we were unable to provide validity data for all measures. Unfortunately, at the time this study was conducted, emerging adult measures did not exist for all the constructs examined; therefore, we utilized adolescent measures for peer relationship self-efficacy and perceptions of social acceptance as the best available proxy.

Although we slightly modified wording to best suit emerging adults and we were able to demonstrate good internal reliability, we acknowledge that the use of measures normed for a different developmental age group is not ideal in that we may not be assessing aspects of social relationships that are less relevant to emerging adulthood. Since conducting the study, a validated emerging adult self-perception measure, *The Self-Perception Profile for College Students*,[51] has been published that would be better to use in future emerging adult research as the scale utilized was only validated in individuals up to 19 years of age.

To our knowledge, there is not yet an emerging adult measure of peer relationship self-efficacy. Strengths of the study include the theoretical framework based on social cognitive therapy and emerging adulthood literature. We applied social cognitive theory by assessing peer and romantic relationship

self-efficacy factors, and peer relationship self-efficacy was identified as critical factors to explore in further research. Future studies may incorporate social cognitive theory more formally to investigate additional constructs (e.g., social support, particularly peer).

Since peer relationship self-efficacy emerged as a key mediator in this study, further research may operationalize peer support as a distinct construct to differentiate between self-efficacy and social support, two constructs of social cognitive theory.

Emerging adulthood is a significant period of personal exploration and growth. Thus, it is imperative that the results of cross-sectional studies among emerging adult childhood cancer survivors are used to test these associations in larger scale multisite studies to develop effective interventions that incorporate psychosocial variables. Psychosocial need assessments are necessary; however, they are not currently provided in a systematic or consistent manner as one study found that only 9% of institutions incorporate standard psychosocial evaluations for childhood cancer survivors.[52] It is necessary for childhood cancer survivors and young adults with cancer to routinely receive systematic assessment of psychosocial health care needs to inform effective interventions to improve perceptions of social acceptance and assure comprehensive care.[52]

Conclusion

Results suggest associations between normative social developmental factors (e.g., parental and peer attachment) and perceptions of social acceptance are mediated, at least in part, by relationship self-efficacy. Although further research is needed to inform a more comprehensive framework incorporating constructs from social cognitive theory and attachment literature, this study provided knowledge by identifying predictors and mediators of perceived social acceptance in the high-need, understudied population of emerging adult cancer survivors. Our findings identified a need for additional research to develop interventions meeting the unique needs of emerging adult survivors of childhood cancers to improve perceptions of social acceptance and reduce associated stereotypes.

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Authors' Contributions

S.L.C.: Formal analysis, funding acquisition, writing—original draft, and writing—review and editing. R.H.F.: formal analysis, funding acquisition, methodology, writing—original draft, and writing—review and editing. A.W.B.: Formal analysis and writing—review and editing. R.D.: Methodology and writing—review and editing. K.G.: Methodology and writing—review and editing. M.S.: Conceptualization, resources, supervision, visualization, methodology, and writing—reviewing and editing.

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Supplementary Material

Supplementary Table S1
Supplementary Table S2

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